

Twice-Told Preferences?

Repeated Elicitation, Response Coding, and Future Household Wealth

Abstract

Repeated elicitation is commonly used to improve the measurement of economic preferences. I test that premise using 56,050 intertemporal-choice responses from 6,808 Japanese households. Responses observed through 2015 predict financial wealth in 2022–2024 conditional on 2016 wealth, allowing a common-sample comparison of one response, averages, response distributions, and split-block IV. Averaging produces a negative displayed-rate gradient, and split-block IV makes it steeper. Yet the gradient is driven by the questionnaire’s 40% endpoint: capping that category makes the slope imprecise, and flexible category models reject displayed-rate linearity. Moreover, post-first responses add virtually no out-of-sample prediction beyond the first category. Repetition therefore makes persistence and coding assumptions visible, but does not produce a demonstrably better predictor of future wealth. The estimates are predictive rather than causal. The result is a boundary condition for treating repeated survey responses as increasingly precise measures of latent preferences.

Keywords: household finance, intertemporal choice, repeated elicitation, response coding, survey measurement.

JEL Codes: D14, D15, D91, G11, G51.

1 Introduction

Repeated survey elicitations are often averaged or used as multiple proxies for a latent preference. The logic is straightforward: if each answer equals a stable signal plus transitory noise, repetition should sharpen the signal. This logic matters in household finance, where preference and belief measures are used to explain wealth and portfolio choices. It can fail, however, when households repeatedly use focal categories, carry stable response styles across waves, or respond to persistent financial circumstances. In those cases, repetition may reproduce the same measurement problem rather than solve it. This paper asks whether repeated elicitation improves the prediction of future household balance sheets or mainly reveals how the elicitation behaves.

The main takeaway is that repetition changes the diagnosis more than the prediction. On the questionnaire's displayed-rate scale, averaging produces a negative prospective wealth gradient and split-block IV makes that gradient steeper. But the result depends on the 40% endpoint, and flexible category models reject the cardinal distances imposed by the displayed rates. Once the response scale is allowed to be flexible, subsequent early-window responses add virtually no predictive content beyond the first category. Repetition can therefore reduce transitory noise while leaving shared

response conventions and misspecified category spacing intact. A steeper repeated-proxy coefficient need not represent a better preference measure.

I establish this result in four steps. First, I use the full response sample to describe persistence and transitions without assigning cardinal distances to the questionnaire labels. Second, I freeze responses in 2010–2015 and compare one response, an average, and split-block IV in predicting 2022–2024 wealth conditional on 2016 wealth. Third, I test the displayed-rate mapping against endpoint recodings and flexible category representations. Finally, I ask whether subsequent early-window responses improve out-of-sample prediction and whether the resulting category profile transports across the two independently sampled panels.

The data come from two national Japanese household panels that ask the same reference person the same intertemporal-choice question as many as fifteen times. The respondent reports how much money thirteen months from now would be required to forgo 10,000 yen one month from now. Answers occupy eight displayed-rate categories from -5% to 40% . From 2010 through 2024, the panels provide 56,050 valid responses from 6,808 households, with nine responses at the median, linked to deposits, securities, and demographic information. The repeated item makes persistence and category transitions observable; the linked balance sheets make its prospective financial content testable.

I first treat the answer as the ordered response that the questionnaire records. Full-sample transition matrices show substantial but incomplete reuse of response labels, with persistence concentrated at salient categories and the upper endpoint. These patterns are informative about

questionnaire use without identifying cardinal distances or labeling every movement as error. I then compare a flexible first-response category, the household's early mean, alternative order-preserving summaries, and the household's early category distribution. For a repeated-proxy sensitivity, I split the 2010–2015 answers into nonoverlapping chronological blocks and instrument either block with the other. Under independent block-error and exclusion restrictions, split-block IV corrects transitory measurement error on the displayed-rate scale. Persistent response style or financial circumstances can enter both blocks, so the estimator remains a measurement sensitivity rather than a causal design.

The outcome design separates the response, baseline, and outcome windows. Responses from 2010–2015 form the proxies, 2016 supplies baseline financial position and controls, and outcomes are averaged over 2022–2024. No baseline or future response enters the early score. This construction removes the look-ahead embedded in a full-panel household mean and asks whether early responses predict financial position six to eight years after baseline. The primary wealth-rank sample contains 3,099 households.

On the displayed-rate scale, a ten-percentage-point higher early mean predicts a 2022–2024 wealth rank 0.64 percentile points lower conditional on 2016 rank. Split-block IV implies 0.94 percentile points, compared with 0.52 for one response. The split-block estimate is detectably steeper than the mean, although the mean and split-block contrasts with the first-response slope are imprecise. Thus repetition and the maintained proxy restrictions change the estimated gradient, but the common scalar comparison does not validate the scale on which that gradient is measured.

The endpoint audit shows why that distinction matters. Recoding the displayed 40% endpoint as 20% makes the negative slope imprecise, while deleting 40% responses flips the slope positive among households with a remaining answer. Saturated models reject displayed-rate linearity using either the first response or repeated response shares. The full category profile rejects monotonicity, but that reversal disappears when the 40% endpoint is excluded; the remaining profile is not theory-aligned decreasing. The negative mapped slope is therefore an endpoint-driven linear projection, not an economically invariant marginal effect.

Flexible representations also clarify what subsequent early-window responses add. Post-first response shares do not add detectably beyond saturated first-response categories, and repeated cross-validation gives virtually identical out-of-sample gains for the first-category and repeated-share representations. Transported category profiles have the same point orientation across panels, but full resampling intervals include zero. Repetition reveals the response distribution, persistence, and proxy-block behavior; it does not produce a clearly superior predictor relative to a flexible first response.

The outcome decomposition supplies a further boundary. Early responses do not detectably predict securities entry, while retention among existing participants is suggestive but imprecise. Attrition corrections leave the wealth-rank estimate nearly unchanged. These checks sharpen the predictive result without eliminating stable confounding by planning ability, cognition, earning capacity, financial literacy, financial circumstances, or response style. The estimates are predictive associations, not causal effects of patience.

The paper contributes by evaluating repetition as a measurement technology rather than assuming its value. Prior work shows that intertemporal-choice measures covary with wealth (Barsky et al. 1997; Epper et al. 2020; Kureishi et al. 2021; Laibson et al. 2026), while multiple-proxy methods show how repeated measures can correct attenuation (Gillen et al. 2019; Beauchamp et al. 2017; Goda et al. 2019; Stango and Zinman 2023; Dohmen and Jagelka 2024). The long panel connects these literatures by holding the item, households, and prospective outcome design fixed while varying how repeated answers are represented. The resulting boundary condition is substantive: correcting transitory error can steepen a coding-dependent coefficient even when the data reject that coding and additional responses do not improve prediction. Persistence, cardinal validity, and predictive gain are separate properties of a repeated measure.

Related work documents temporal instability, persistent preference shocks, and person-specific response noise (Meier and Sprenger 2015; Chuang and Schechter 2015; Salamanca 2018; Arts et al. 2026; Belzil and Jagelka 2025). Money-based intertemporal responses can also combine time preference with utility curvature, liquidity, outside returns, attention, and response conventions (Cubitt and Read 2007; Dean and Sautmann 2021; Fulford and Shupe 2026; Belot et al. 2025); household-finance choices reflect rich preference heterogeneity and portfolio frictions (Calvet et al. 2026; Choukhmane and de Silva 2026). I therefore use the neutral term *elicited required return* and do not interpret the response as a structural discount parameter. The broader implication is methodological: with coarse repeated survey measures, averaging and multiple-proxy correction should not be assumed to dominate a flexible first-response encoding.

Section 2 describes the panels, elicitation, outcomes, and prospective design. Section 3 compares the predictive content of one response and repeated representations, then uses repetition to diagnose scale use and persistence. Section 4 locates the result across household-finance margins, Section 5 evaluates robustness and interpretation, and Section 6 concludes.

2 Data and Prospective Design

This section introduces the two household panels, the repeated elicitation, and the prospective outcome design. The key timing choice is to freeze all response information by 2015, condition on the household's 2016 balance sheet, and measure later outcomes in 2022–2024. The design therefore compares alternative representations of exactly the same early answers rather than allowing later responses to enter the predictor.

2.1 Panels and elicitation

The data come from the Japan Household Panel Survey (JHPS) and the Keio Household Panel Survey (KHPS), administered by the Panel Data Research Center at Keio University. KHPS began in 2004 and JHPS in 2009; common questionnaires have been administered jointly since 2014. Each follows a national, population-based sample of Japanese adults and their households and records deposits, securities, income, expenditure, employment, and demographic characteristics. The samples are distinct. I prefix household identifiers by survey, never link respondents across

surveys, and include survey indicators in pooled specifications.

The analysis begins in 2010, the first year of the repeated item in JHPS; KHPS contributes the same item from 2014. Through 2024, the pooled measurement sample contains 56,050 valid responses from 6,808 households. A household contributes 8.23 answers on average, nine at the median, and as many as fifteen. The empirical asset is repeated observation of the same item and reference person, linked to the household balance sheet.

The questionnaire asks respondents how much money thirteen months from now they would require instead of receiving 10,000 yen one month from now. They choose one of eight amounts displayed with annual rates of -5 , 0 , 2 , 4 , 6 , 10 , 20 , and 40 percent. A higher category records a higher required return to wait. Both payment dates are in the future, so this is not an immediate-versus-delayed present-bias task.

The response scale is economically meaningful but not cardinally identified by the labels. Categories are unevenly spaced, and answers cluster at 0 , 10 , 20 , and 40 percent; the two most common categories account for more than half of all responses. Displayed rates provide a transparent benchmark for economic magnitudes, but they impose distances that an ordered response alone does not identify. I therefore compare that benchmark with flexible category representations and order-preserving scalar encodings.

The answer can combine willingness to delay consumption with utility curvature, liquidity, outside returns, attention, misunderstanding, and a convention for using the response scale (Andersen et al. 2008; Andreoni and Sprenger 2012; Cohen et al. 2020; Dean and Sautmann 2021; Fulford and

Table 1. Sample construction and response distribution

<i>Panel A. Response categories</i>									
Displayed annual rate	−5%	0%	2%		4%	6%	10%	20%	40%
Share of responses (%)	0.3	15.3	4.7		4.2	6.2	27.8	16.8	24.6
<i>Panel B. Repetition and prospective sample</i>									
Valid elicitation-year observations									56,050
Households with a valid elicitation									6,808
Elicitations per household: mean / median / IQR / maximum							8.23 / 9 / 4–11 / 15		
At least two valid responses in 2010–2015									6,096
Early score and a 2016 household record									4,831
Early score, 2016 wealth rank, and complete controls									4,716
Primary later-wealth-rank sample									3,099

Notes: Panel A uses all valid JHPS and KHPS responses from 2010–2024. Panel B reports household counts. The prospective design uses 2010–2015 responses, a 2016 baseline, and outcomes measured in 2022–2024.

Shupe 2026). I use the neutral term *elicited required return*. The paper studies the measurement and predictive content of that response, not a directly observed structural discount factor.

2.2 Financial outcomes

The primary outcome is financial-wealth rank. Financial wealth equals deposits plus securities and excludes housing, pensions, other real assets, and debt. I calculate midranks within survey and calendar year before imposing the prospective sample, so the outcome measures a household’s position in its contemporaneous wealth distribution and is invariant to currency units, skewness, and common nominal changes. The Internet Appendix reports complete variable construction, missing-component rules, summary statistics, and a specification requiring both balance-sheet components.

Three complementary outcomes show where a rank association comes from. Poisson pseudo-

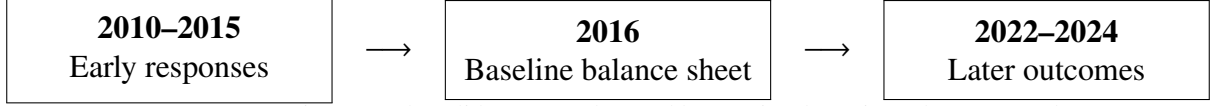
maximum likelihood retains zeros and gives a denomination-invariant level specification. A two-part model separates the probability of positive wealth from log wealth conditional on being positive. Log-one-plus wealth is retained only as a unit-sensitive check because roughly one-quarter of observed household-years report zero financial wealth.

Securities participation equals one when the household reports positive stocks, bonds, or investment trusts. The prospective outcome is the share of available 2022–2024 waves with participation. I also distinguish entry among 2016 nonparticipants from retention among 2016 participants. This split prevents a pooled coefficient from hiding offsetting transitions.

2.3 Timing and specification

Figure 1 summarizes the design. For each household I freeze the response information set at 2015, measure baseline outcomes and controls in 2016, and average observed outcomes over 2022–2024. No baseline or future response enters the early score. The design removes the mechanical look-ahead in a full-panel household mean and asks whether early responses predict financial position six to eight years after baseline.

Figure 1. Prospective timing



Notes: Response summaries use only valid answers observed by 2015 and require at least two early responses. Baseline outcomes and controls are measured in 2016. Later outcomes average all available observations from 2022 through 2024.

Let q_i^{pre} denote the mean displayed rate among household i 's valid 2010–2015 responses, requiring at least two. For a linear outcome j , the benchmark specification is

$$y_{ij}^{2022:24} = \alpha_j + \beta_j q_i^{\text{pre}} + \theta_j y_{ij}^{2016} + Z_i^{2016'} \delta_j + \lambda_{s(i)} + e_{ij}, \quad (1)$$

where Z_i^{2016} contains age and age squared, sex, marital status, and the number of early responses, and $\lambda_{s(i)}$ is a survey fixed effect. Participation specifications also condition on baseline wealth rank. The PPML wealth-level model uses the same covariates with a log link. I report heteroskedasticity-robust standard errors; the split-block estimator clusters by household.

For each linear outcome, I compare three representations on exactly the same outcome-specific sample. *First response* uses only the earliest valid 2010–2015 answer. *Early mean* uses all eligible answers. For *split-block IV*, I order early answers chronologically and form nonoverlapping odd- and even-position means,

$$r_i^o = s_i + u_i^o, \quad r_i^e = s_i + u_i^e. \quad (2)$$

I stack the two instrument directions, using each block as an instrument for the other, include a proxy-block indicator, and cluster by household (Gillen et al. 2019). The estimator isolates the

association with the component common to both blocks if the blocks measure the same persistent component, their block-specific errors are conditionally independent, and those errors are unrelated to the later-outcome residual. The common component can still contain persistent response style, planning ability, or financial circumstances. Split-block IV is therefore an assumption-dependent measurement sensitivity, not a causal design.

The primary wealth-rank sample contains 3,099 households. It is a selected long-horizon sample, and baseline conditioning changes the estimand from a cross-sectional level gap to conditional future financial position. Section 5 evaluates survey heterogeneity, attrition, weights, and alternative baselines. The estimates remain predictive rather than causal.

3 What Repetition Adds

The central comparison holds the households, timing, controls, and outcome fixed while changing only how the early responses are represented. Repetition makes the displayed-rate projection steeper under the split-block restrictions, but it does not deliver a detectably better rank predictor than the first response. Its clearest return is instead diagnostic: repeated answers reveal the consequences of response coding, focal scale use, and persistence.

Table 2. Repeated elicitation and later household finances

Outcome	First response	Early mean	Split-block IV	Dep. mean	R^2	N
Wealth rank (0–100)	−5.24*** (1.89)	−6.36*** (2.45)	−9.40*** (3.61)	49.6	0.678	3,099
Wealth level (PPML)	−0.242** (0.113)	−0.282* (0.145)	–	1,263	–	3,099
Positive-wealth incidence	−0.032 (0.037)	−0.069 (0.048)	−0.102 (0.070)	0.744	0.454	3,099
Conditional log wealth	−0.226 (0.139)	−0.348** (0.174)	−0.509** (0.253)	6.600	0.448	2,498
Securities participation	−0.043 (0.033)	−0.054 (0.042)	−0.080 (0.062)	0.243	0.496	3,054
Entry (2016 nonholders)	−0.008 (0.030)	−0.011 (0.039)	−0.017 (0.059)	0.082	0.047	2,284
Retention (2016 holders)	−0.179* (0.106)	−0.244* (0.128)	−0.334* (0.173)	0.719	0.097	770

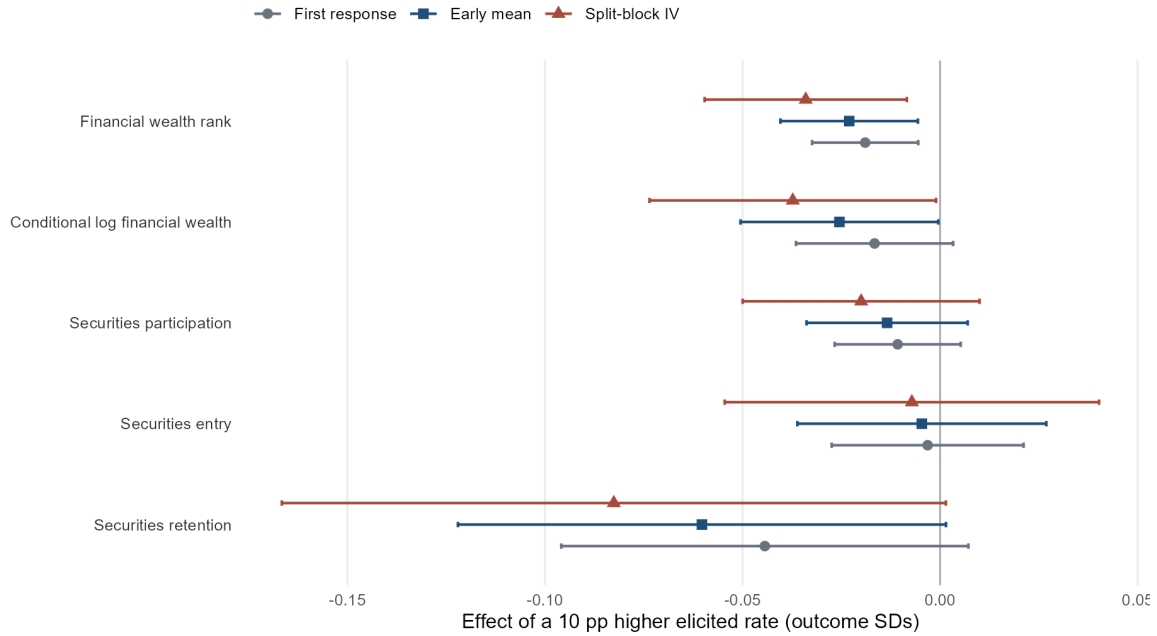
Notes: Rate-score coefficients are per unit (100 percentage points); multiplying by 0.1 gives a ten-percentage-point contrast. The first response and early mean use heteroskedasticity-robust standard errors. Symmetric split-block IV stacks nonoverlapping odd/even early-response proxies and clusters by household. All rows condition on the corresponding 2016 baseline outcome, age and age squared, sex, marital status, early-response count, and survey fixed effects; participation also conditions on baseline wealth rank. The dependent-variable mean and R^2 are from each row’s early-mean estimation sample and specification; the wealth-level mean is in units of 10,000 yen. PPML is nonlinear, so no split-block IV entry or OLS R^2 is reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.1 Benchmark predictive comparison

On the common 3,099-household sample, a ten-percentage-point higher first response predicts a 0.52-percentile lower wealth rank in 2022–2024. The corresponding early-mean and split-block estimates are 0.64 and 0.94 percentiles. A paired household bootstrap distinguishes split-block IV from the mean, but neither repeated-response estimate is statistically distinguishable from the first-response slope for wealth rank. Repetition therefore steepens the displayed-rate point estimate under the block assumptions without establishing a superior predictor.

Table 2 extends the comparison across household-finance outcomes. Figure 2 translates the linear estimates into outcome-standard-deviation units. The negative gradient is concentrated in financial wealth: the point estimates for participation and entry are small, while retention among existing participants is more suggestive but less precise.

Figure 2. Predictive gradients across measurement choices



Notes: Points report the effect of a ten-percentage-point higher elicited required return in outcome-standard-deviation units; bars are 95% confidence intervals. All three estimators use the same outcome-specific repeat-eligible sample and the controls in equation (1). First response and early mean use heteroskedasticity-robust standard errors. Split-block IV uses nonoverlapping early-response blocks and household-clustered standard errors.

The comparison does not by itself show that repetition has recovered a better measure. Averaging can reduce transitory variation, while split-block IV can correct attenuation if its proxy restrictions hold. Both operations can also preserve a shared response convention or a misspecified cardinal mapping. The next two exercises use the repeated panel to separate those possibilities.

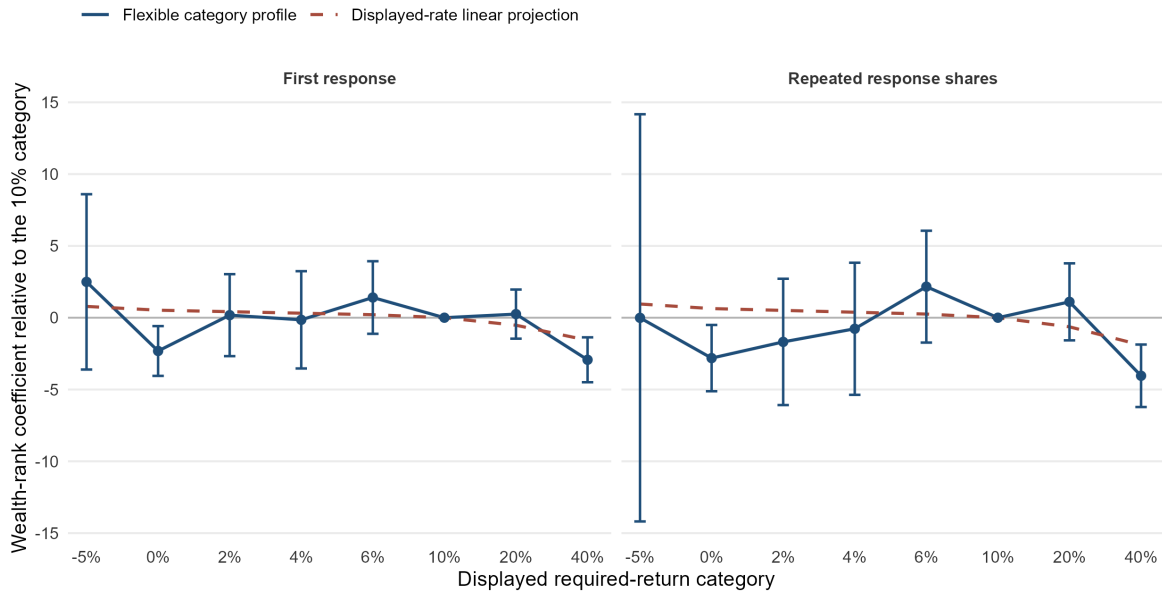
3.2 Coding the response scale

The questionnaire's 40% endpoint drives the negative displayed-rate slope. It represents 25.3% of eligible early answers, and 44.9% of households use it at least once. Recoding every 40% answer as

20% on the same 3,099 households changes the early-mean coefficient from -6.36 to -2.91 and makes it imprecise. Deleting 40% responses before recomputing the mean flips the slope positive among the 2,780 households with a remaining answer. The latter changes the sample and is therefore an influence diagnostic, not the same estimand.

Saturated category models reject the linear distances imposed by the displayed rates. The restriction is rejected for both the first response, $F(6, 3084) = 3.43$ ($p = 0.0023$), and repeated response shares, $F(6, 3084) = 3.80$ ($p = 0.0009$). The full repeated-share profile rejects monotonicity in either direction. Without the 40% endpoint, the reversal disappears, but the remaining profile is compatible with a weakly increasing—not a theory-aligned decreasing—pattern. Figure 3 shows why: the flexible estimates do not trace the smooth negative profile implied by the displayed-rate regression, and the upper endpoint is discontinuous with the adjacent category. The Internet Appendix reports the full endpoint audit, restriction tests, and bootstrap procedure.

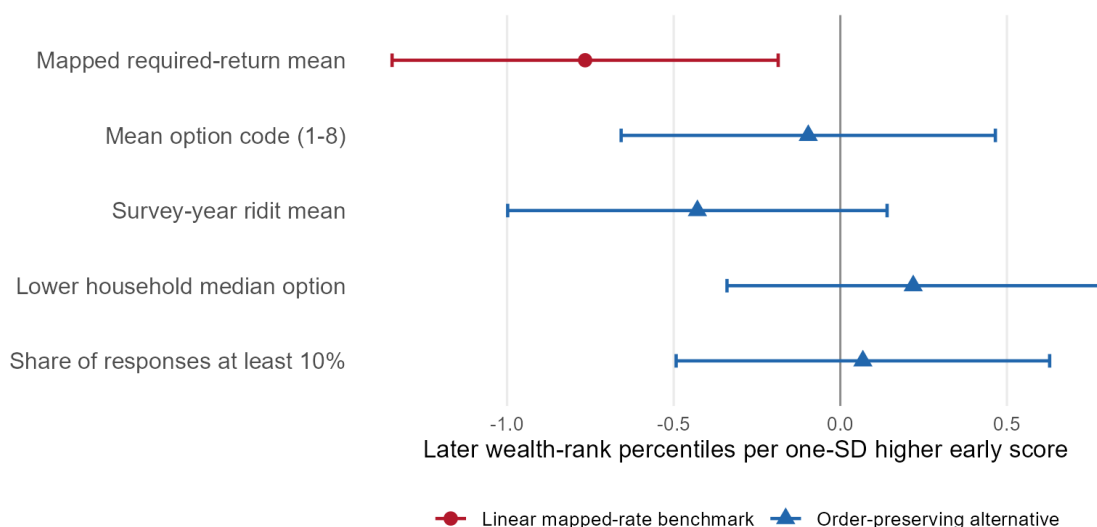
Figure 3. Flexible category profiles and the displayed-rate projection



Notes: Points are flexible category coefficients relative to the 10% response, with 95% heteroskedasticity-robust confidence intervals. The left panel uses saturated first-response categories; the right panel uses household shares in the eight early-response categories. Dashed lines show the category profile implied by a linear regression on the displayed rates. All models use the common wealth-rank sample and the controls in equation (1).

Order-preserving scalar summaries reinforce the distinction between ordinal information and the questionnaire's cardinal distances. Figure 4 compares the mapped early mean with the mean option code, survey-year rudit, household median category, and share of responses at or above 10%. Each score is standardized on the same sample. The four ordinal alternatives are individually imprecise, even though the displayed-rate mean is not. This difference in individual significance is not itself a formal test; the saturated restrictions above provide that test.

Figure 4. Later wealth rank under alternative response encodings



Notes: Points report later wealth-rank percentiles per one-standard-deviation increase in each 2010–2015 response summary; bars are 95% confidence intervals. All specifications use the same 3,099 households, condition on 2016 wealth rank and the controls in equation (1), and orient higher scores toward a higher required return.

Once the first response is represented flexibly, subsequent early-window responses add little prediction. Seven post-first category shares are jointly insignificant conditional on saturated first-response dummies, $F(7, 3077) = 1.02$ ($p = 0.418$), with partial $R^2 = 0.0022$. In repeated survey-stratified cross-validation, incremental out-of-sample R^2 relative to baseline controls is 0.00525 for first-response dummies and 0.00534 for repeated shares; the mapped mean yields 0.00185. Training the flexible share profile in one panel and applying it to the other gives the same point orientation in both directions, but two-sample bootstrap intervals include zero. Repetition exposes category composition and scale failure without producing a demonstrably better predictor than a flexible first category.

3.3 Persistence and questionnaire use

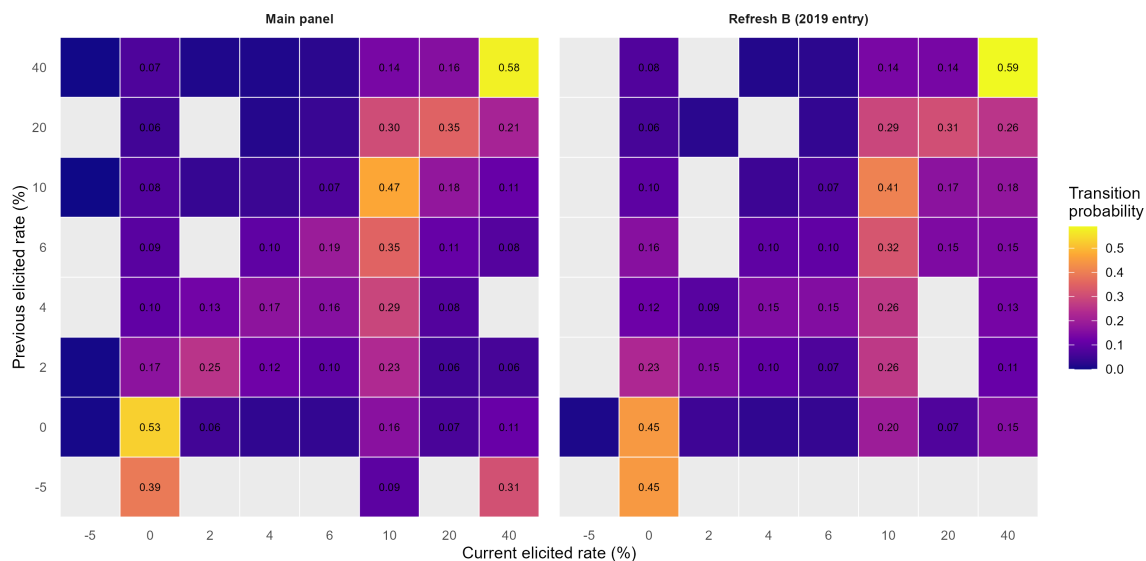
Repeated answers also reveal how households use the scale. Let $Y_{it} \in \{1, \dots, 8\}$ denote the recorded option in year t , and define the row-normalized transition matrix

$$T_{jk} = \Pr(Y_{it} = k \mid Y_{i,t-1} = j).$$

Across all 56,050 responses, the adjacent-category Spearman correlation is 0.513 and the exact-repeat rate is 0.444. Conditional on a response of 0, 10, 20, or 40 percent, the next answer remains in that salient set with probability 0.893. The 40% endpoint retains 0.578 of its occupants from one year to the next. These are properties of questionnaire use; they do not label every movement as error or validate the distances between labels.

Figure 5 compares annual transitions in the main panel with the JHPS cohort that entered in 2019. Both matrices show a diagonal, but movements also concentrate at salient categories. The refresh cohort's exact-repeat rate is 0.415 and focal-to-focal retention is 0.896, close to the corresponding main-panel values. Similar focal retention among newer respondents makes a convention unique to long-tenured households less plausible, although cohort composition and a common questionnaire prevent a causal panel-conditioning interpretation.

Figure 5. Observed annual transitions in elicited-rate categories



Notes: Cells are row-normalized probabilities of the current category conditional on the previous category. The left panel uses the 56,050-response main sample; the right panel uses the JHPS cohort that entered in 2019. Cells supported by fewer than ten adjacent pairs, plus a complementary cell when required for disclosure protection, are suppressed. The figure contains no fitted probabilities.

The distinction is the answer to the paper's title question. Repetition can reveal persistence, focal responding, and whether nonoverlapping proxy blocks move together. It cannot, by itself, distinguish a persistent preference from a persistent economic state or response style. Nor can persistence validate a cardinal scale that the category data reject.

4 Household-Finance Margins

The benchmark comparison establishes a prospective relation with financial-wealth rank. This section asks where that relation appears in the balance sheet. The evidence is consistent across a rank outcome, a level model, and the positive-wealth amount, but it does not extend to securities

entry. The result is therefore about later financial position, not a general explanation for portfolio participation.

4.1 Financial wealth

The early-mean coefficient for later wealth rank is -6.36 , so a ten-percentage-point higher elicited required return predicts a rank 0.64 percentile points lower conditional on 2016 rank. The early score's standard deviation is 12.05 percentage points, making the corresponding one-standard-deviation contrast 0.77 percentile points. Baseline rank absorbs much of the persistent cross-sectional wealth gap, so this magnitude describes relative position six to eight years later rather than wealth inequality accumulated before 2016.

The level and two-part specifications locate the association. Poisson pseudo-maximum likelihood implies 2.8% lower expected financial wealth for a ten-percentage-point higher early mean; its confidence interval narrowly includes zero. The same contrast predicts a 0.69-percentage-point lower probability of positive wealth and 0.035 lower log wealth conditional on being positive, with the latter more precise. Applying the conditional-positive point estimate to the relevant positive-wealth median of about eight million yen gives roughly 275,000 yen. This translation is illustrative: the estimate becomes imprecise when the sample is restricted to households with positive wealth at baseline.

Log-one-plus wealth has the same sign but is not a denomination-free percentage specification when zeros are present. Expressing the same balance sheet in yen instead of units of 10,000 yen

changes the coefficient because the added constant does not scale with the units. The Internet Appendix reports that sensitivity, inverse-hyperbolic-sine estimates, strict component-observation rules, and the full two-part results. I treat the sign agreement across rank, PPML, and the two-part model as the robust evidence rather than interpreting their transformation-specific magnitudes as interchangeable.

4.2 Participation, entry, and retention

The pooled participation gradient is small and imprecise. A ten-percentage-point higher early mean predicts a 0.54-percentage-point lower share of later waves with securities participation; split-block IV implies 0.80 percentage points. Both intervals include zero.

Separating transitions shows that the data contain no detectable entry margin. Among 2,284 households without securities in 2016, the early-mean estimate is -0.11 percentage points per ten percentage points, with a 95% interval from approximately -0.9 to $+0.7$ percentage points. Among 770 baseline participants, the corresponding retention estimate is -2.44 percentage points, and split-block IV implies -3.34 points. The retention estimates are suggestive but imprecise in a smaller sample.

This boundary is economically plausible. Securities participation depends on fixed costs, defaults, attention, trust, and access as well as preferences (Gomes et al. 2021; Choukhmane and de Silva 2026). An elicited required return can predict later liquid-wealth position without being the margin that determines whether a household opens an investment account.

4.3 Baseline and timing checks

The rank gradient is stable when the baseline is averaged over several years. Replacing the single 2016 outcome with averages over 2013–2016, 2014–2016, or 2015–2016 yields ten-percentage-point effects between -0.59 and -0.65 percentile points. A spline in 2016 rank and survey-specific nuisance profiles also leave the mapped projection nearly unchanged. These specifications address baseline functional form and regression-to-the-mean concerns, although the longer baseline windows overlap the response period.

A common two-response window removes the difference between the nearly six early answers available in JHPS and the two available in KHPS. Using only 2014–2015 responses gives a pooled early-mean effect of -0.58 percentile points per ten percentage points and a split-block effect of -0.88 percentile points. A separately timed JHPS design—2010 baseline rank, 2011–2012 responses, and 2013–2015 outcome ranks—produces an effect of -0.84 percentile points. This is an earlier prospective replication, not a pretrend placebo, and both exercises inherit the displayed-rate limitation.

The prospective gradient is smaller than its contemporaneous counterpart because the designs answer different questions. Equation (1) conditions on the earlier balance sheet, uses an auditable long-horizon survivor sample, and excludes responses collected alongside or after the outcome. A pooled contemporaneous regression can combine historical accumulation, current financial circumstances, and future-score leakage. The prospective design sacrifices power to remove the last channel and make the baseline gap explicit; it does not identify a causal accumulation effect.

5 Robustness and Interpretation

Three checks matter for interpretation. The wealth gradient has the same sign in two independently sampled panels, observed attrition and alternative weighting do not overturn it, and auxiliary income, financial-literacy, and saving evidence is consistent with an accumulation channel. These checks strengthen the predictive interpretation without identifying a causal preference effect.

5.1 Two independently sampled panels

JHPS and KHPS use the same item but provide different measurement intensity. Eligible JHPS households contribute 5.89 early answers on average and have an odd/even block correlation of 0.70. KHPS households contribute exactly two answers, with a correlation of 0.41. This contrast asks whether the pooled result depends on the unusually long JHPS response history.

The early-mean wealth-rank coefficients are -9.27 in JHPS and -4.92 in KHPS. Both have the same sign, and their equality-test p -value is 0.329, although the KHPS estimate is individually imprecise. Survey-specific split-block coefficients are also negative: -11.33 in JHPS and -8.15 in KHPS, with first-stage F statistics of 1,966 and 380. The common two-wave design in Section 4 points in the same direction. I view this as directional replication across distinct household samples, not external validation across questionnaires, countries, or institutions.

The transition comparison in Figure 5 supplies a complementary tenure check. Focal-category retention is nearly identical in the main panel and the 2019 refresh cohort even though exact repetition is somewhat higher among long-tenured respondents. Repeated exposure may still change

recall, attention, or scale use (Cernat et al. 2026); the cohort comparison is descriptive rather than randomized.

5.2 Attrition and weighting

Later wealth is observed for 3,099 of 4,716 households that satisfy the pre-outcome requirements, a retention rate of 65.7%. Conditional on the 2016 information set, a ten-percentage-point higher early response predicts only a 0.55-percentage-point lower probability of observing later wealth, with a standard error of 0.56 percentage point. The analogous participation estimate is similarly small and imprecise.

Inverse-probability weights based only on early responses, baseline outcomes, demographics, response count, and survey change the rank coefficient from -6.36 to -6.22 . Official 2016 cross-sectional weights preserve its sign but reduce precision. Requiring two or all three later wealth observations produces slightly larger negative estimates. These exercises address selection on observed baseline variables and balanced follow-up; they cannot recover households selected out by unobserved future shocks.

5.3 Construct-validity evidence

Income and financial-literacy checks leave a residual wealth gradient. In the income-complete JHPS prospective sample, adding observed 2016 household income changes the same-sample mapped coefficient from -9.80 to -10.34 . Contemporaneous specifications with income and a financial-

literacy battery show the same qualitative pattern. These controls are incomplete and sometimes observed after the earliest elicitation, so they narrow rather than eliminate stable-confounding interpretations.

Four JHPS waves with both household income and expenditure provide an auxiliary saving-rate result. A ten-percentage-point higher response is associated with a 0.64-percentage-point lower saving rate using one response, 1.37 points using the household mean, and 1.64 points using a lagged-response instrument. This pattern is consistent with an accumulation channel, but it is based on noisy self-reports from 2010–2013 and remains observational.

The outcome decomposition points in the same direction: later wealth rank, PPML wealth, and conditional positive wealth move together, while securities entry does not. This is consistent with more patient or financially organized households accumulating larger liquid balances. It is also consistent with persistent cognition, planning, earning capacity, resource access, or reporting style affecting both the elicitation and the balance sheet.

5.4 Scope of the evidence

The split-block instrument corrects transitory proxy error only under its maintained restrictions; it is not an exogenous shock to intertemporal preference. A strong first stage rules out weak-instrument instability, not correlated block errors or stable confounding. Likewise, response persistence does not validate the displayed-rate distances, and a prospective association does not recover a structural discount factor.

Outcome measurement creates an additional boundary. Experimental choice inconsistency is more strongly associated with self-reported than administrative wealth in recent evidence, suggesting that decision quality may also predict reporting error (Arts et al. 2026). JHPS and KHPS do not contain an administrative balance-sheet benchmark. Wealth ranks, PPML, the two-part decomposition, and baseline conditioning reduce dependence on any single transformation, but they cannot eliminate error shared across survey measures.

The robust conclusion is therefore narrower and more useful than a preference-effect claim. Early response categories contain information about later financial wealth; repeated elicitation exposes persistence and scale dependence; and neither averaging nor split-block IV delivers a demonstrably better predictor once the first answer is encoded flexibly.

6 Conclusion

Repeated elicitation can make a survey coefficient larger without making the underlying measure more valid or more predictive. In two Japanese household panels, required-return responses are persistent, averaging produces a negative prospective wealth gradient, and split-block IV makes that displayed-rate gradient steeper. But the result depends on how the questionnaire's top category is coded, and flexible category models reject the linear displayed-rate scale.

Once the response scale is treated flexibly, most of the predictive content is already present in the first category. Subsequent early-window responses reveal transitions, focal-category use, and shared

proxy behavior, but add virtually no out-of-sample prediction. The main return to repetition in this application is therefore diagnostic: it exposes assumptions that a one-shot elicitation leaves hidden.

This conclusion is a boundary condition, not a claim that repetition is generally unhelpful. Repetition should be most valuable when errors are transitory and proxy-specific; it will be less valuable when coding conventions or persistent traits are shared across waves. For other coarse preference and belief measures, researchers should benchmark repeated summaries against a flexible first-response model and validate category spacing before interpreting averaging or proxy IV as improved measurement. The estimates here are predictive and do not identify a structural discount factor or a causal effect of patience.

Data Availability

JHPS and KHPS microdata are distributed under a research-use agreement by the Panel Data Research Center at Keio University and cannot be redistributed. Researchers can apply to the Center. The replication code reconstructs the analysis panel from licensed files and regenerates all tracked aggregate tables and figures. No respondent identifiers or restricted microdata are included.

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